

Solution Requirements - ShopEZ Stock Trading Platform

Parameter	Details
Date	18 June 2026
Team ID	SE-09
Project Name	ShopEZ Stock Trading Platform
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Profiles	- Form-based Registration with role selection (Trader, Admin). - Login with JWT token creation and browser-storage persistence. - Profile header displaying username, email, and virtual cash balance.
FR-2	Live Market Tracker	- Tabular stock display: Ticker, Name, Current Price, Change (%). - Background simulated price updates occurring every few seconds. - Dynamic visual alerts (mint-green/soft-red) when prices tick.
FR-3	Ticker Analysis & Charting	- Interactive page for selected stock showing real-time price charts (using Chart.js in light theme). - Statistics display: Open, High, Low, Volume, and Change. - Trade form embedded directly on details page.
FR-4	Virtual Trade Execution	- Form input validating integer share count. - Buy order validation: user has sufficient cash balance. - Sell order validation: user owns sufficient shares. - Real-time update to User balance and Transaction ledger.
FR-5	Portfolio Tracker	- Dynamic portfolio display showing average purchase cost per stock. - Current valuation and total profit/loss calculated on live prices. - Visual circular charts highlighting asset distribution.
FR-6	Transaction Ledger	- Tabular transaction logs: Trans ID, Ticker, Type (BUY/SELL), Shares, Price, Timestamp. - Filters to view logs by ticker symbol or date.
FR-7	Platform Moderation Portal	- Admin dashboard summary showing total users, active listings, transaction volume. - Add new stock listing with initial starting price. - Delete user accounts or delete stock listings from the system.

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.